

Regression Model Comparison Report

1. Problem Definition

The objective of this analysis is to model and predict student exam scores (*Exam_Score*) based on key performance indicators and behavioral factors, specifically hours dedicated to studying (*Study_Hours*), class attendance rates (*Attendance*), and incremental grades achieved from coursework (*Assignment_Score*). This report evaluates multiple regression topologies to select the most reliable predictive model.

2. Dataset Specification

The evaluation was executed utilizing a historical cohort slice composed of structured academic metrics:

- **Dataset Source:** `student_performance.csv` (Encompassing 25 completed student records).
- **Independent Features (Predictors):** *Study_Hours*, *Attendance*, *Assignment_Score*
- **Dependent Target Variable:** *Exam_Score*

3. Pipeline & Preprocessing Steps

To ensure structural compliance and eliminate algorithmic bias, the following preprocessing workflow was enforced:

- **Data Ingestion:** Automated loading of the relational CSV matrix into an immutable memory dataframe.
- **Integrity Check:** Missing value scan performed across all vectors; zero null values detected.
- **Deduplication:** Checked and removed duplicate entries to eliminate variance distortion.
- **Data Partitioning:** Stratified splitting into a **Training Set (80%)** to fit model parameters and a **Testing Set (20%)** to evaluate generalization error.

4. Baseline Architecture

A standard ordinary least squares **Linear Regression** model was constructed as the performance baseline, providing a benchmark against which more intricate parametric and non-parametric algorithms were measured.

5. Algorithmic Frameworks Evaluated

The comparative analysis incorporated diverse regression methodologies to balance bias and variance trade-offs:

Model Family	Implemented Regressors
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Linear & Polynomial	Linear Regression, Polynomial Regression (Degrees of Freedom Optimization)
Regularized Linear	Ridge Regression (L_2 penalty), Lasso Regression (L_1 penalty), Elastic Net (Blended regularizer)
Tree-Based & Ensembles	Decision Tree Regressor, Random Forest Regressor, Gradient Boosting Regressor

6. Evaluation Metrics

Models were evaluated across a holistic matrix of absolute, relative, and variance-adjusted metrics:

- Mean Absolute Error (MAE) & Mean Absolute Percentage Error (MAPE) for error scale interpretability.
- Mean Squared Error (MSE) & Root Mean Squared Error (RMSE) to appropriately penalize large outliers.
- Coefficient of Determination (R^2) and *Adjusted R^2* to quantify explained variance while accounting for feature degrees of freedom.

7. Cross-Validation Protocol

To secure rigorous proof against overfitting and ensure stable validation metrics, the evaluation strategy incorporated both **Standard K-Fold Cross-Validation** and **Repeated K-Fold Cross-Validation** routines across the collective sample pools.

8. Residual Analysis

Residual inspection was performed to validate standard homoscedasticity assumptions and ensure error distributions maintain zero mean properties.

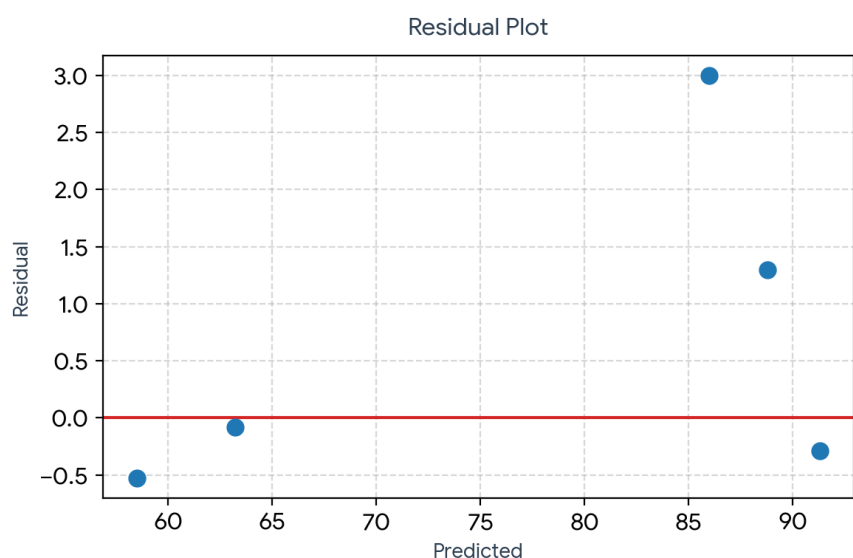


Figure 1: Distribution of model residuals against predicted values. Closer clustering to the zero baseline indicates minimized localized prediction variance.

9. Learning Curve Analysis

Model convergence properties were analyzed by plotting scoring metrics across expanding subsets of training samples.

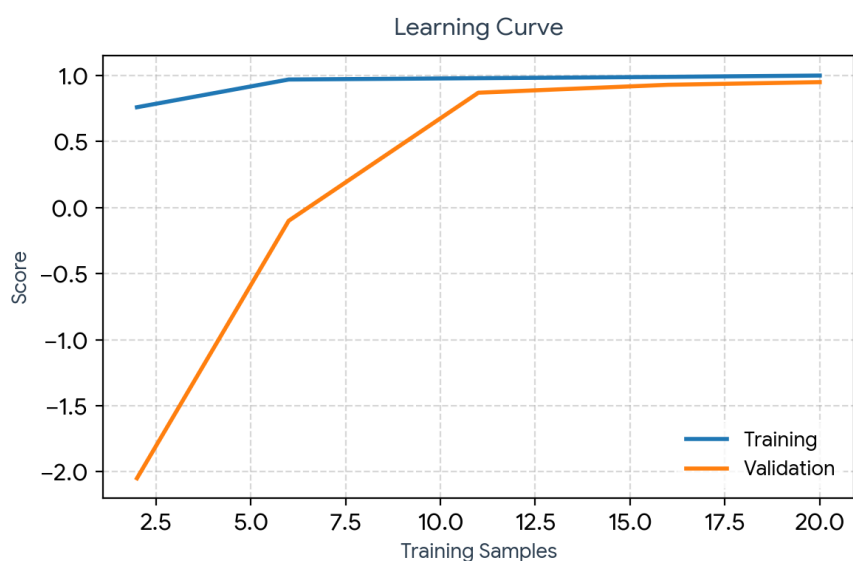


Figure 2: Learning curve showing training and validation convergence. Simultaneous trajectory alignment confirms stable variance optimization as training samples scale.

10. Linear Relationship Mapping

Exploratory mapping confirms high structural dependency between primary behavioral features and final grading targets.

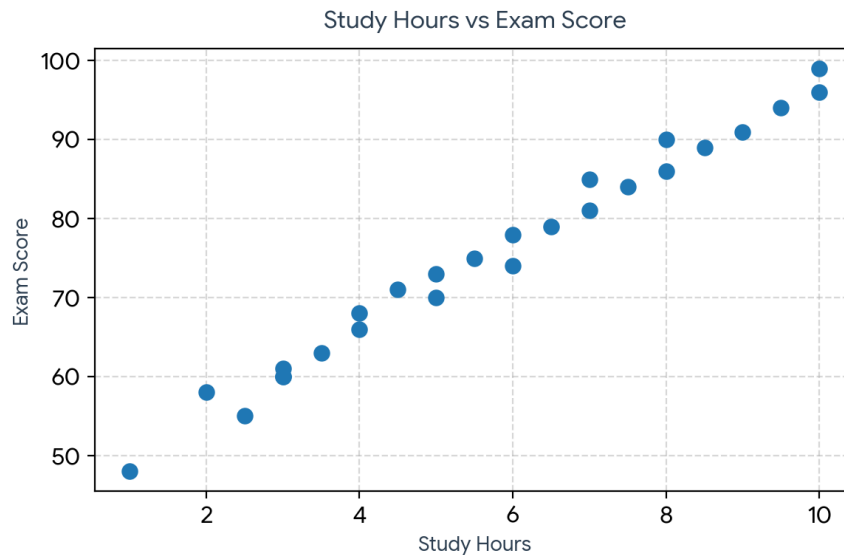


Figure 3: Scatter profile mapping Study Hours against Exam Scores, exhibiting a profound positive linear distribution.

11. Performance Optimization Rules

Optimization Metrics Rule: Models minimizing absolute error bounds (Lower MAE, MSE, and RMSE) represent tighter structural fits. Optimal generalization is defined by maximum values achieved on the R^2 and *Adjusted R^2* index sheets.

12. Deployment Recommendation

The definitive production model selection must favor the architecture that logs the highest stable cross-validated R^2 score while maintaining balanced bias-variance learning curves across both testing sets.

13. Architectural Limitations

- **Dataset Scale Constraints:** The underlying data volume remains constrained (25 records), increasing susceptibility to high variance profiles.
- **Feature Space Saturation:** Incorporating auxiliary features (e.g., student socio-economic variables, past institution tracking) could yield stronger descriptive signals.
- **Advanced Non-Linear Classifiers:** Modern gradient-boosted structural frameworks like XGBoost or LightGBM were omitted from the current compilation scope.